CS-1101
PROGRAMMING FUNDAMENTALS

Lab 11

By

Engr. Muhammad Awais

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Lab Contents

- Functions

Lab Objectives

- The objective of this lab is to understand the concept of functions in C++.

Functions

- A function is a block of code that performs a specific task.
- The idea is to put some commonly or repeatedly done tasks together to make a function so that instead of writing the same code again and again for different inputs, we can call this function.
- A function is a block of code that runs only when it is called.
- Syntax to declare a function

```
returnType functionName (parameter1, parameter2,...) {
    // function body
}
```

Functions Cont….

- Function Declaration.

```
// function declaration
void greet() {
    cout << "Hello World";
}
```

- Function Calling.

```
int main() {

    // calling a function
    greet();

}
```

How function works

```
#include<iostream>

void greet() {
    // code
}

int main() {
    ... .. ...
    greet();
    ... .. ...
}
```

function call

Function Parameters

```
#include<iostream>

void displayNum(int n1, double n2) {
    // code
}

int main() {
    ... ...
    displayNum(num1, num2);
    ... ...
}
```

function call

Return Statement in a Function

```
#include<iostream>

int add(int a, int b) {
    return (a + b);
}

int main() {
    int sum;

    sum = add(100, 78);
    ... ...
}
```

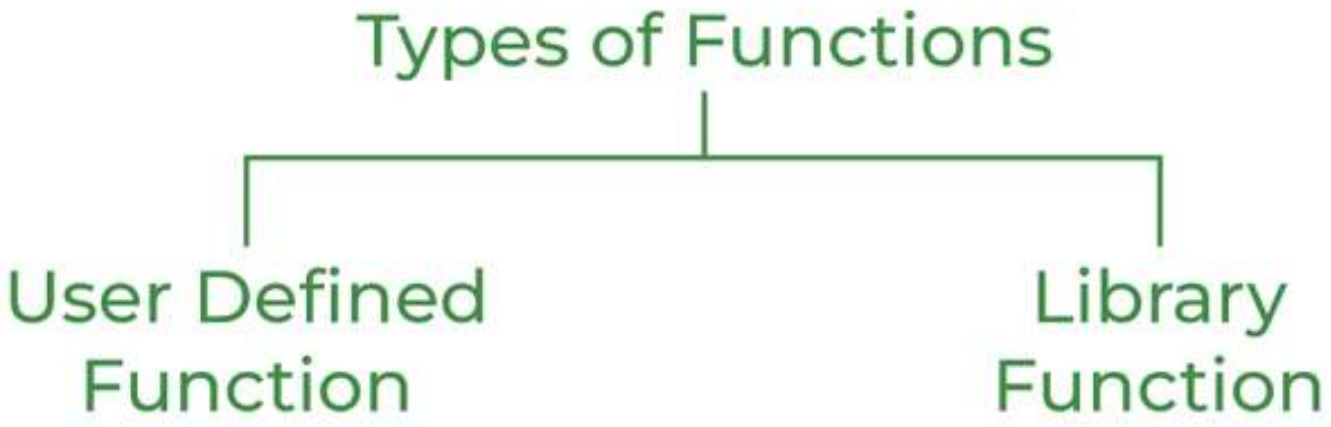
function call

Function Prototyping

```
#include<iostream>
using namespace std;
//function prototyping
void greet();

int main(){
//function calling
    greet();
    return 0;
}
//function definition
void greet(){
    cout << "Hello! Good Morning" << endl;
}
```

Types of Functions



Tasks of Library Functions

- Write a C++ program that uses library function to find out the following for a number input by user.
  - Square Root
  - Absolute of a number
  - Sin
  - Cos
  - Power of 3.

Tasks of User Defined Functions

- **Function with Void Return Type (Greeting Message):**

Write a C++ program that defines a function to display a welcome message. The function should not take any parameters and should print "**Hello, Welcome to C++ Functions!**" when called in the main() function.

- **Function with If-Else Statement (Even or Odd):**

Write a C++ program that defines a function to determine whether a given number is even or odd. The function should take an integer input and print "Even" if the number is divisible by 2; otherwise, it should print "Odd". Call this function from main() with different values.

- **Function with a Loop (Factorial Calculation)**:

Write a C++ program that defines a function to compute the factorial of a given number. The function should use a loop to multiply all numbers from 1 to n and return the result. In main(), prompt the user to enter a number and display the calculated factorial.

Tasks of Functions

- **Menu-Driven Program - Calculator:**

Write a C++ program that functions as a simple calculator. The program should:

- Display a menu with basic arithmetic operations:
  - Addition, Subtraction, Multiplication, and Division.
- Ask the user to select an operation.
- Prompt the user to enter two numbers.
- Use separate functions to perform each arithmetic operation.
- Display the result of the selected operation.
- Allow the user to perform multiple calculations until they choose to exit.

Tasks of Functions

- **Menu-Driven Program – Restaurant Bill:**

Write a C++ program to calculate the total bill for a restaurant order. The program should:

- Display a menu with at least five food items and their prices.
- Allow the user to select an item by entering its corresponding number.
- Ask the user to enter the quantity for the selected item.
- Use a function to calculate the total cost of the selected item (total = price × quantity).
- Allow the user to order multiple items until they choose to exit.
- Use another function to calculate and display the final bill, including a service charge or tax (e.g., 10%).
- Display the total amount before exiting.

Where to Use Functions?

- A function to display the menu.
- A function to calculate the cost of an item (total = price × quantity).
- A function to calculate the final bill with taxes/service charges.

References

- https://www.geeksforgeeks.org/functions-in-cpp/